DR. OSCAR MENDEZ MALDONADO

Director of AI & Data Science · Visiting Associate Professor in Robotics and AI

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EXPERIENCE

Director of AI & Data Science — Locus Robotics
2025 – Present · Remote, Boston HQ

- Grew the AI and Data Science team roughly threefold, and lead it.
- Designed and delivered System Directed Labor: orchestrating humans, robots, and AI as a unified operational system to optimise warehouse throughput at fleet scale.
- Defined and executed Locus Robotics' customer- and investor-facing AI strategy, translating complex technical capabilities into clear, credible narratives for enterprise and board-level audiences.
- Led execution on high-impact AI initiatives including foundation models, LLM integration, and fleet-level intelligence across the deployed robot fleet.
- Designed and deployed MLOps capabilities across Locus Robotics' global customer sites.
- Inventor on a pending patent for warehouse labour orchestration.

Machine Vision Lead — Locus Robotics
2023 – 2025 · Remote, Boston HQ

- Designed the Computer Vision pillar at Locus Robotics and hired a dedicated team to own perception across the robot fleet.
- Built proof-of-concept foundation models for vision, spanning depth estimation, semantic segmentation, and visual odometry.
- Ran on-robot trials for localisation and segmentation on edge devices, liaised cross-functionally to deploy validated models into production.
- Contributed to multiple open-source libraries, including pose graph optimisation tools used across the robotics community.

(Visiting) Associate Professor in Robotics and AI — University of Surrey
2021 – Present · Guildford, Surrey

- Lead research in Robotics, AI and Computer Vision at CVSSP, supervising researchers and managing multi-disciplinary projects.
- Projects span embodied intelligence, autonomous navigation, and perception: Agritech, Robot Guide Dog, Medical Assistive Robots, BEV Estimation using GNNs, BEV-SLAM.
- Core technical contributor across Transformers, GNNs, CNNs, ROS, SLAM, 3D Reconstruction, NERFs, C/C++, Python, Pytorch, OpenCV.
- Recognised "Resident ROS Expert"; inventor on four granted US patents in sensor calibration, segmentation and BEV semantic mapping.

Research Fellow I/II — CVSSP
2017 – 2021 · Guildford, Surrey

- Led independent research in Computer Vision and Robotics, from fundamental algorithms to deployable systems. Project managed industrial and publicly funded research; supervised PG researchers.
- Research areas: Localisation using Heatmap Regression, AV Navigation, Robot Grasping, Segmentation, 3D Reconstruction, Visual and LiDAR SLAM.
- Created and delivered ISO-standard ready localisation technology for multi-storey car parks and bespoke AV software on NVIDIA PX2.

Engineering Intern — Qualcomm Ltd.
2011–2012 · Farnborough, Hampshire

ABOUT ME

I am an award-winning AI researcher and strategy leader with a PhD in Artificial Intelligence, specialising in robotics and computer vision. I bridge fundamental research with real-world deployment, helping organisations turn AI and autonomous robotics into scalable, reliable systems. In industry, I lead AI strategy for autonomous mobile robots. In academia, I contribute to perception, autonomous systems, and trustworthy AI.

AWARDS & HONOURS

ICRA 2022 Best Paper Award
Translating Images into Maps

Sullivan Thesis Prize 2018
Best Computer Vision Thesis in the UK

4 US Patents
Sensor calibration, segmentation and BEV mapping; warehouse labour orchestration pending

40+ Publications & Patents
750+ citations, h-index 13

BMVA Executive Committee
Sponsorship Chair for UK Computer Vision Association

Professional Memberships
FHEA · IEEE Member · IET Associate

EXPERTISE

AI Strategy · Team Building & Hiring · Research Management · Grant Writing · AI Governance & Ethics · Foundation Models & LLMs · Computer Vision · 3D Perception & SfM · SLAM & Localisation · Robotics & ROS · MLOps (Kubernetes, MLflow, DVC) · Python & PyTorch · C/C++

EDUCATION

PhD Robotics, Computer Vision & AI
CVSSP
Guildford, Surrey
2013 – 2017 · Award Winning Thesis

BEng Electronic and Computer Engineering (Hons)
University of Surrey
Guildford, Surrey
2009 – 2013 · First Class (Hons) Degree

SELECTED PROJECTS

System Directed Labor
2025 – Pres. NDA

Agentic AI, *Logistics*, *Robotics*, *Reinforcement Learning*, *Multi-Agent Systems*, *MLOps* Technical Team Director, Co-Project Manager. Designed and delivered System Directed Labor: orchestrating humans, robots, and AI as a unified agentic system to optimise warehouse throughput at fleet scale (100-1000s of People+Robots).

FlexBot
2021 – 2025 £1.0M

Agritech, *Robotics*, *Computer Vision*, *Navigation* Principal Investigator. Acquired funding and project managed programme until I left university.

Precision Agriculture using Hyperspectral Imagery
2021 – 2026 £144,000

Hyperspectral Imagery, *Contrastive Learning*, *Mapping*, *Remote Sensing* Principal Investigator, Acquired and Project Managed a research project on Precision agriculture using Drones and Satellite Imagery

Autonomous Valet Parking
2018 – 2020 £1.2M

Mapping, *Localisation*, *SLAM*, *Autonomous Vehicles*, *Sensor Fusion* Wrote, delivered and Project Managed an InnovateUK Project in collaboration with Parkopedia Ltd., Streetdrone Ltd. and Transport Systems Catapult. Represented the University of Surrey in all matters related to AVP, including reporting, planning and delivery.

SELECTED PUBLICATIONS

HYDRA: HYbrid knowledge Distillation and spectral Reconstruction Algorithm
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. C Thirgood, **O Mendez**, E Ling, J Storey, S Hadfield

The Radiance of Neural Fields: Democratizing Photorealistic and Dynamic Robotic Simulation
ICRA, 2025. G Nuthall, R Bowden, **O Mendez**

SpaGBOL: Spatial-Graph-Based Orientated Localisation
WACV, 2025. T Shore, **O Mendez**, S Hadfield

PEnG: Pose-Enhanced Geo-Localisation
RA-L, 2025. T Shore, **O Mendez**, S Hadfield

BEV-CV: Birds-Eye-View Transform for Cross-View Geo-Localisation
IROS 2024. T Shore, S Hadfield, **O Mendez**

Two Hands Are Better Than One: Resolving Hand to Hand Intersections via Occupancy Networks
IEEE FG, 2024. M Ivashechkin, **O Mendez**, R Bowden

Utilizing GNNs for Adverse Health Detection in Sensor-Based Remote Monitoring
Computers in Biology and Medicine, 2024.
N Bijlani, **O Mendez**, R Nilforooshan, P Barnaghi, S Kouchaki

Learning Adaptive Neighborhoods for Graph Neural Networks
ICCV, 2023. A Saha, **O Mendez**, C Russell, R Bowden

Translating Images into Maps
ICRA, 2022.
A Saha, **O Mendez**, C Russell, R Bowden Best Paper Award

"The Pedestrian next to the Lamppost" Adaptive Object Graphs for Better Instantaneous Mapping
CVPR, 2022. A Saha, **O Mendez**, C Russell, R Bowden

BEV-SLAM: Building a Globally-Consistent WorldMap Using Monocular Vision
IROS, 2022.
J Ross, **O Mendez**, A Saha, M Johnson, R Bowden US Patent

SeDAR: Reading Floorplans Like a Human – Using Deep Learning to Enable Human-Inspired Localisation
IJCV, 2020. **O Mendez**, S Hadfield, N Pugeault, R Bowden

Taking the Scenic Route to 3D: Optimising Reconstruction From Moving Cameras
ICCV, 2017. **O Mendez**, S Hadfield, N Pugeault, R Bowden

SELECTED RESEARCH

Localisation and Mapping using Raman Spectrometry
2022–Present Christopher Thirgood

NERF-based Mapping for Robot Guide Dog
2022–Present Georgina Alcolado Nuthall

Cross-View Geolocalisation using BEV Estimation
2022–Present Tavis Shore

SLAM and BEV Mapping for Autonomous Maritime Vehicles
2020–2024 Dr. James Ross

Instantaneous Mapping and Graph-Based Learning
2019–2023 Dr. Avishkar Saha

Visual Odometry for Autonomous Vehicle Stability
2019 – 2022 Dr. Nimet Kaygusuz

Precision Agriculture using Hyperspectral Sensors
2022–Present Dr. Daniel Ayuba

Anomaly Detection using GNNs
2021–2025 Dr. Nivedita Bijlani

Body Pose Estimation using Kinematics and Vision
2022–2026 Dr. Maksym Ivashechkin

Visual Navigation using Reinforcement Learning
2019–2023 Dr. Xihan Bian

Autonomous Calibration and Planning for Autonomous Vehicles
2017–2021 Dr. Celyn Walters

HOBBIES

Basketball | DIY | Video Games | Weightlifting | Building Robots

LANGUAGES

English
Native Level

Spanish
Native Speaker

PERSONAL PROJECTS

HypoBot

Developed a robot capable of delivering insulin to diabetic patients with mobility issues. Managed PG students and developed project management skills.